Economic Struggles

A relationship between GDP and unemployment

Datathon 2026

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Is there a disparity between how GDP affects men and women's unemployment rate?

Analysis

Variables

Gdp_per_capita_usd
female_unemployment_pct
male_unemployment_pct

Why compare men and women's unemployment rate?

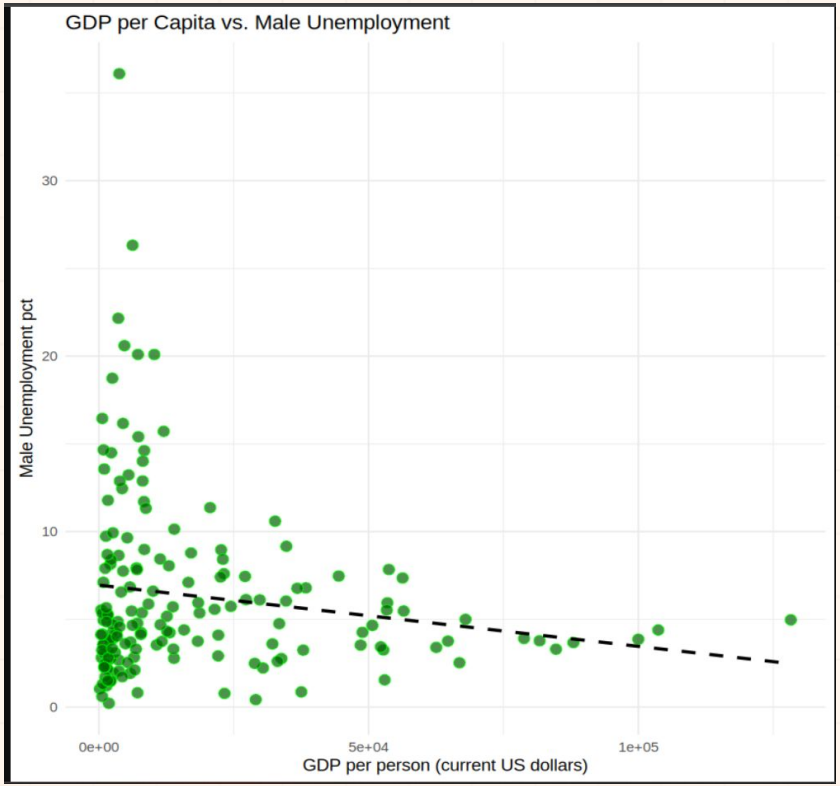
Men and women tend to cluster in different sectors (domains of work)

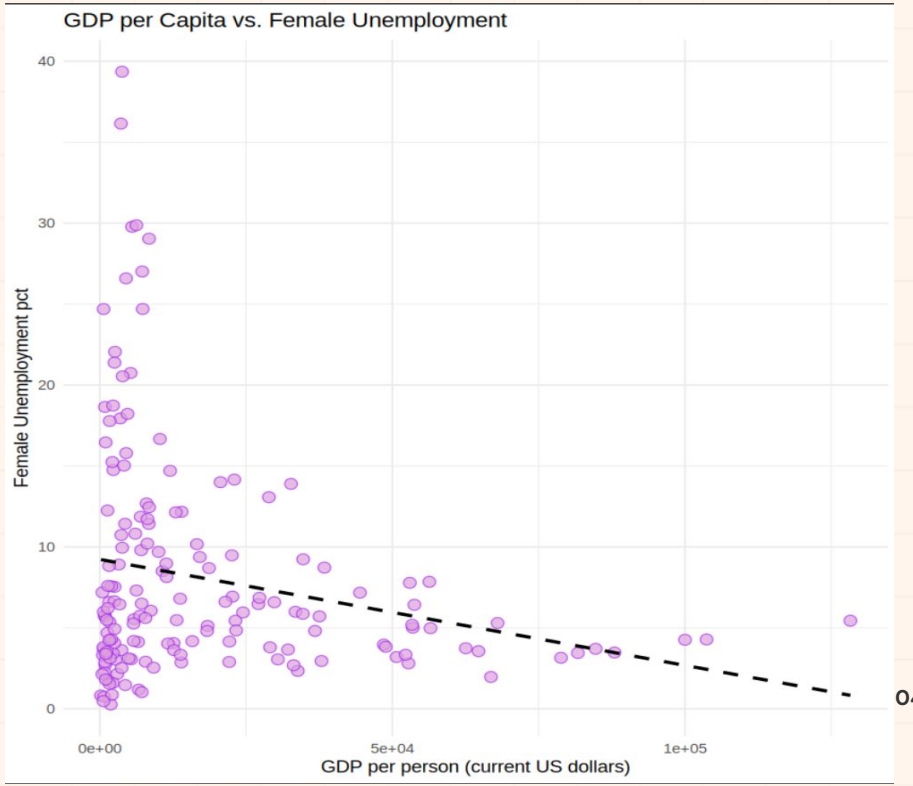
Factors like gender, race, and disability can show systemic disadvantages

```
1] 168
 sum (!keep)
1] 49
 summary(pair[, needed])
gdp_per_capita_usd female_unemployment_pct
Min.   :   199.6   Min.   : 0.266
1st Qu.:  2263.9   1st Qu.: 3.520
Median :  7163.4   Median : 5.658
Mean   : 17660.9   Mean   : 8.072
3rd Qu.: 23212.1   3rd Qu.:10.004
Max.   :128259.4   Max.   :39.354
 cor(pair[[x_name]], pair[[y_name]])
1] -0.2174002
```

```
[1] 168
> sum (!keep)
[1] 49
> summary(pair[, needed])
 gdp_per_capita_usd male_unemployment_pct
 Min.   :   199.6   Min.   : 0.215
 1st Qu.:  2263.9   1st Qu.: 3.253
 Median :  7163.4   Median : 4.732
 Mean   : 17660.9   Mean   : 6.336
 3rd Qu.: 23212.1   3rd Qu.: 7.858
 Max.   :128259.4   Max.   :36.098
> cor(pair[[x_name]], pair[[y_name]])
[1] -0.1586066
>
```

Analysis


GDP per Capita vs. Male Unemployment
Male Unemployment pct
0
10
20
30
0e+00
5e+04
1e+05
GDP per person (current US dollars)


GDP per Capita vs. Female Unemployment
Female Unemployment pct
0
10
20
30
40
0e+00
5e+04
1e+05
GDP per person (current US dollars)

Data Insights/Conclusion

Gender

Female Unemployment is slightly higher, but mainly affected by outliers

- Mean for female unemployment: 8.072
- Mean for male unemployment: 6.34

Unemployment is right skewed which suggests unusually high unemployment rates.

Overall

When GDP is lower, both men and women unemployment percentage is high. Alternatively when GDP is higher, both men and women unemployment percentage is low.

Some low-GDP countries have near 0% unemployment, while others spike very high (some near or above 20-30%).

Further Thoughts

The difference between male and female unemployment is small and does not support the idea that a country's GDP alone affects women and men differently.

What other factors, like work domains/career fields could contribute to the small gender gap?

How does this trend change over time?